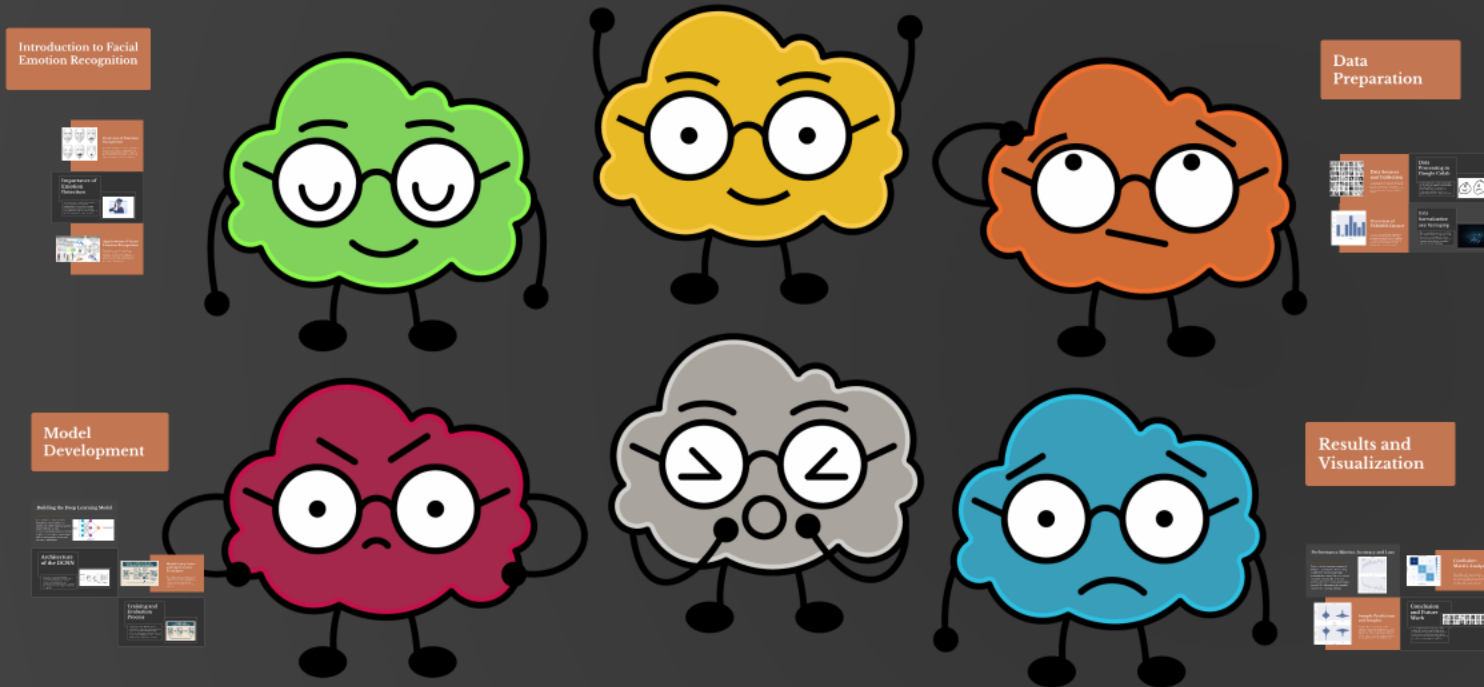


# Facial Emotion Recognition

Teaching machines to read human emotions, one face at a time.



Presented by:  
Sai Bharath Goud Pudari  
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spudari@kent.edu

# Introduction to Facial Emotion Recognition



## Overview of Emotion Recognition

Emotion recognition is the process of identifying human emotions from facial expressions. It's grounded in psychological research indicating that emotions such as happiness, sadness, and anger manifest distinctly on facial features.

## Importance of Emotion Detection

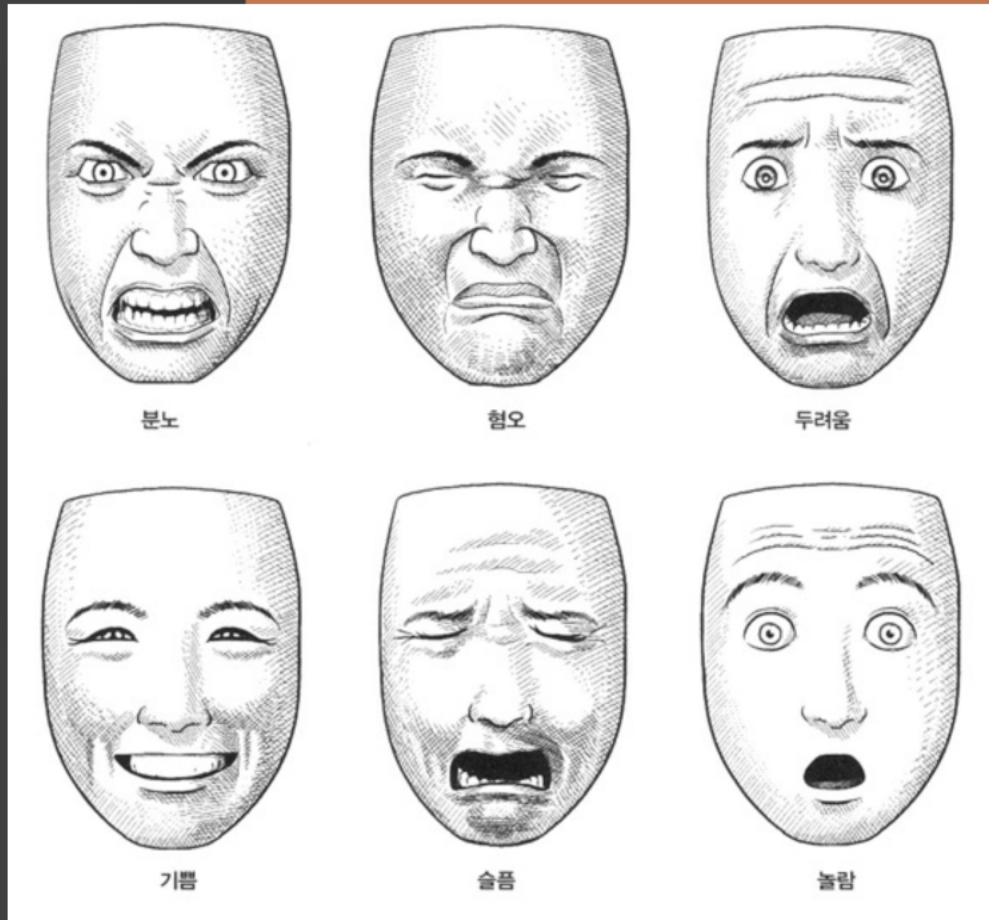
Detecting emotions through facial expressions is critical for enhancing interpersonal communications in various fields, including psychology and customer service. Accurate emotion detection can improve user experiences and facilitate empathy in virtual interactions.



## Applications of Facial Emotion Recognition

Facial emotion recognition has diverse applications ranging from mental health diagnosis to enhancing user engagement in applications and games. It is also utilized in marketing to gauge customer responses to products and advertisements.



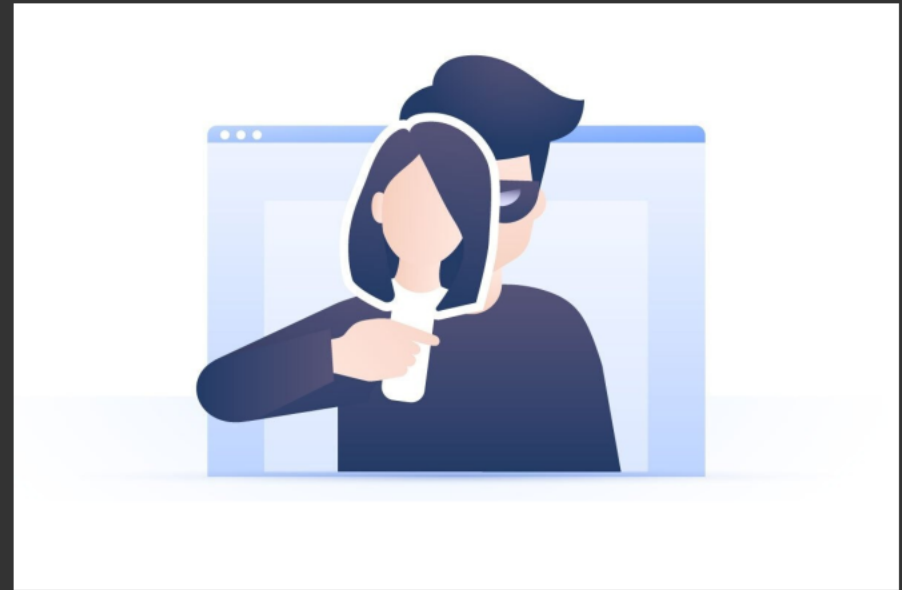


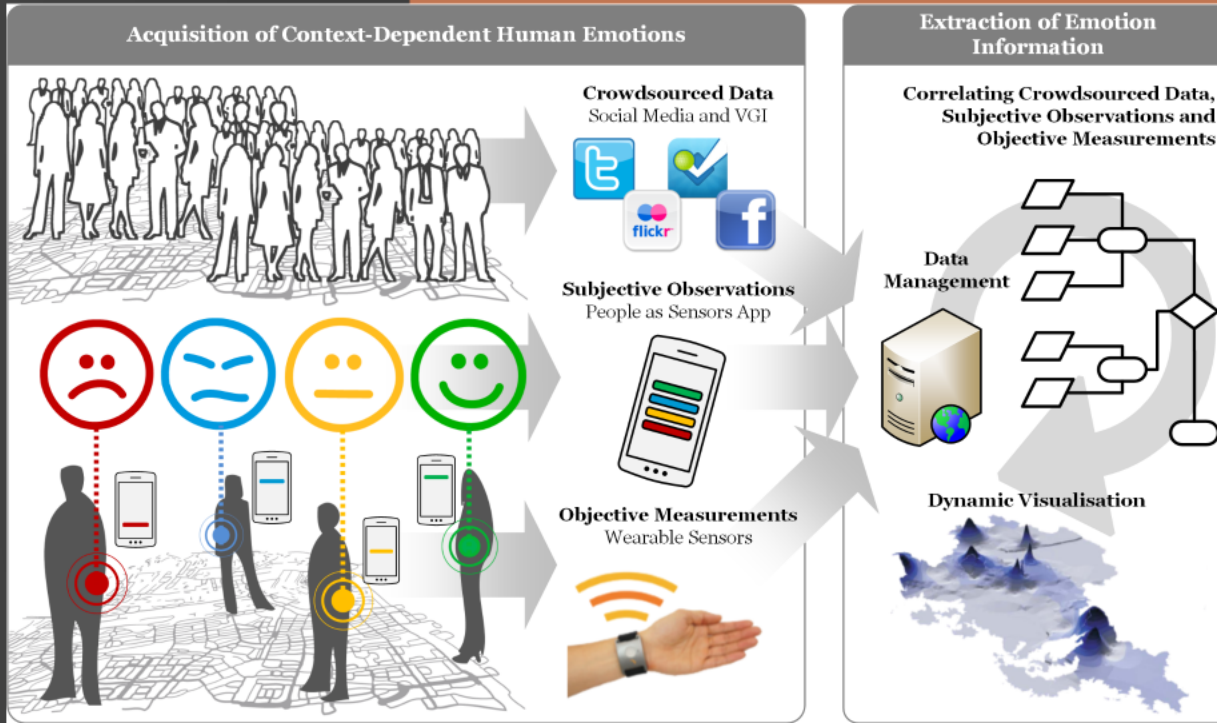
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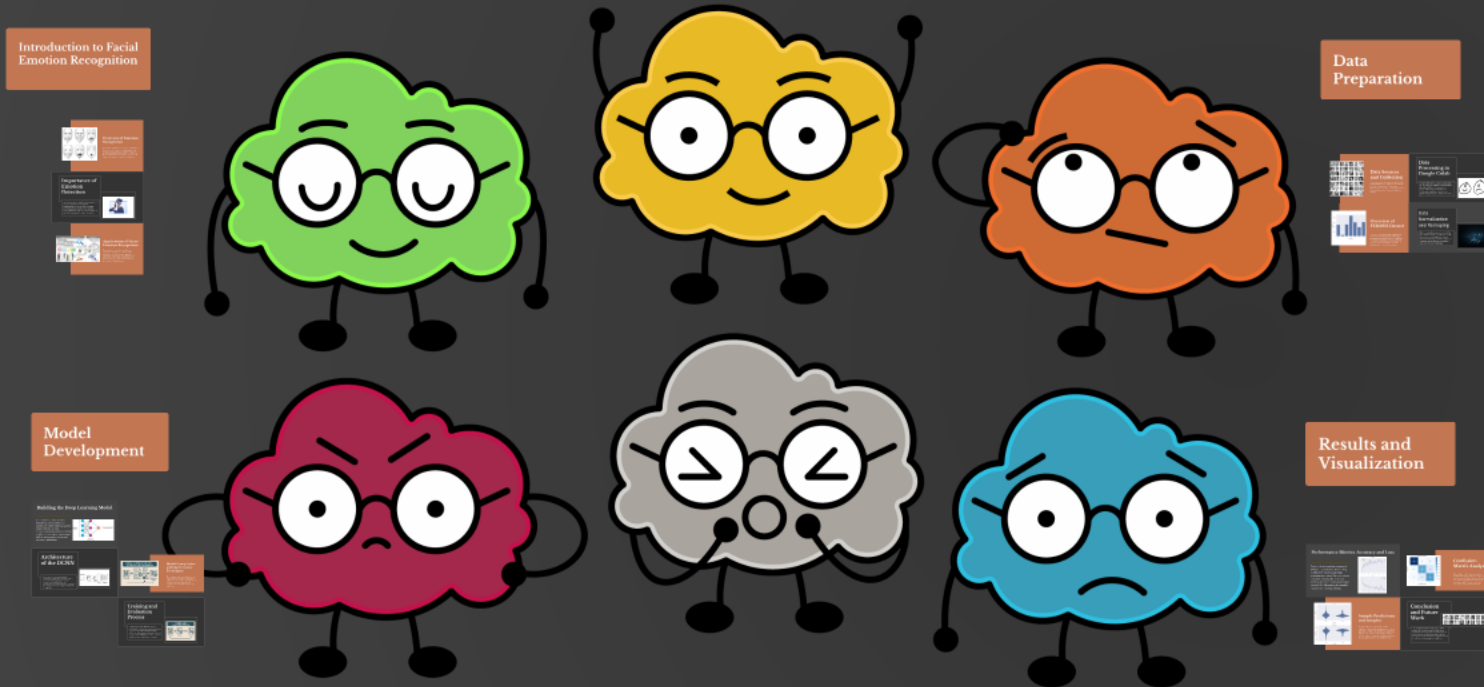


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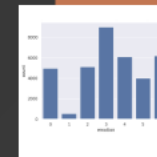
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# Data Preparation



## Data Sources and Collection

The primary source for emotion recognition data is the FER2013 dataset, which comprises 35,887 grayscale images of faces. These images represent various emotions, facilitating extensive training and evaluation for emotion detection models.



## Overview of FER2013 Dataset

The FER2013 dataset includes images labeled with seven different emotions: anger, disgust, fear, happiness, sadness, surprise, and neutral. The diverse representation of these emotions enhances model training efficiency and effectiveness in real-world applications.

## Data Processing in Google Colab

Google Colab serves as an excellent platform for processing large datasets due to its cloud-based capabilities. Functions such as importing libraries, data loading, and conducting critical data analysis can be performed seamlessly in Colab's environment.



## Data Normalization and Reshaping

Data normalization is crucial to ensure that pixel values of images range between 0 and 1, enhancing the learning process of neural networks. Reshaping two-color images to a consistent format, typically square pixels, is critical for effective model input.



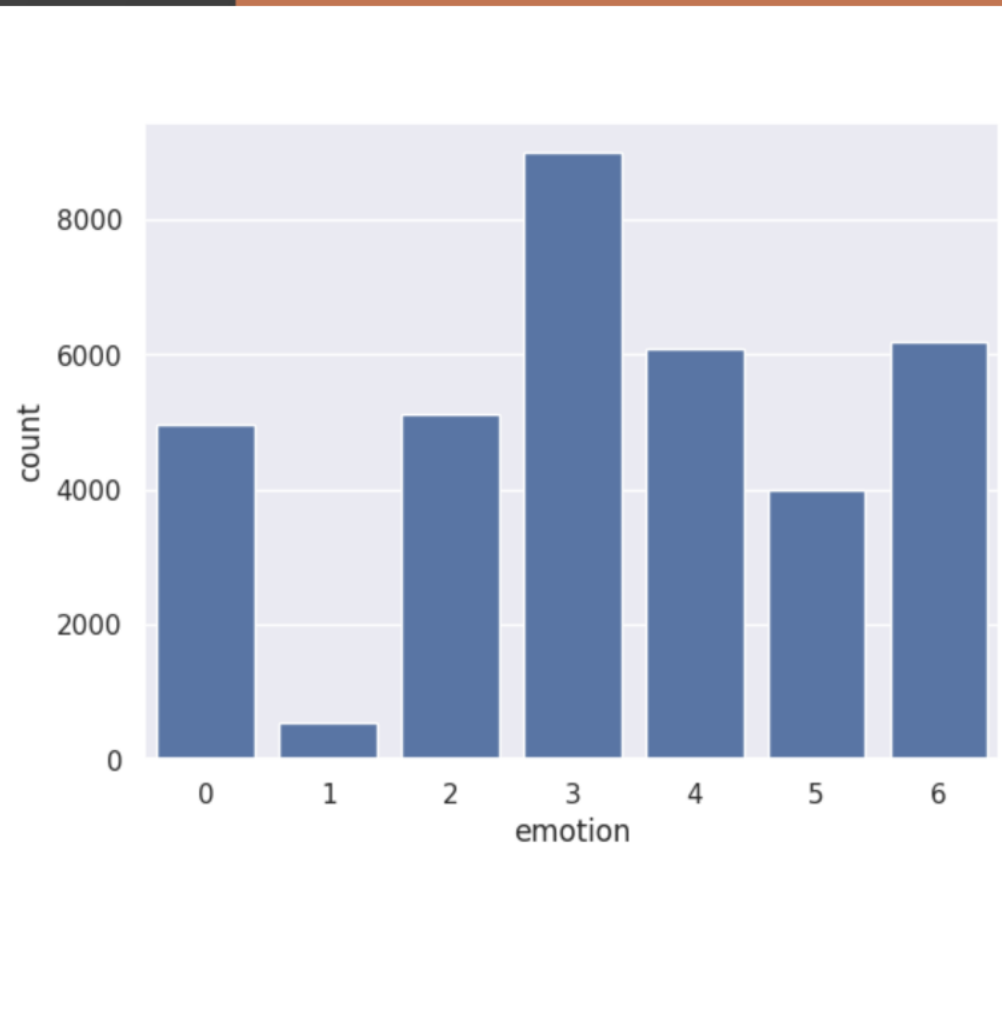




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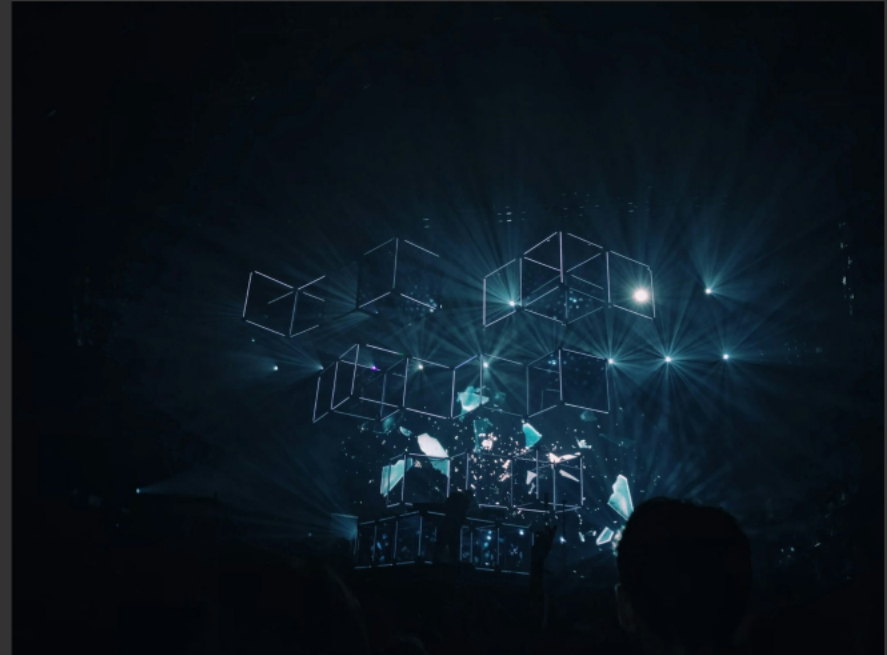


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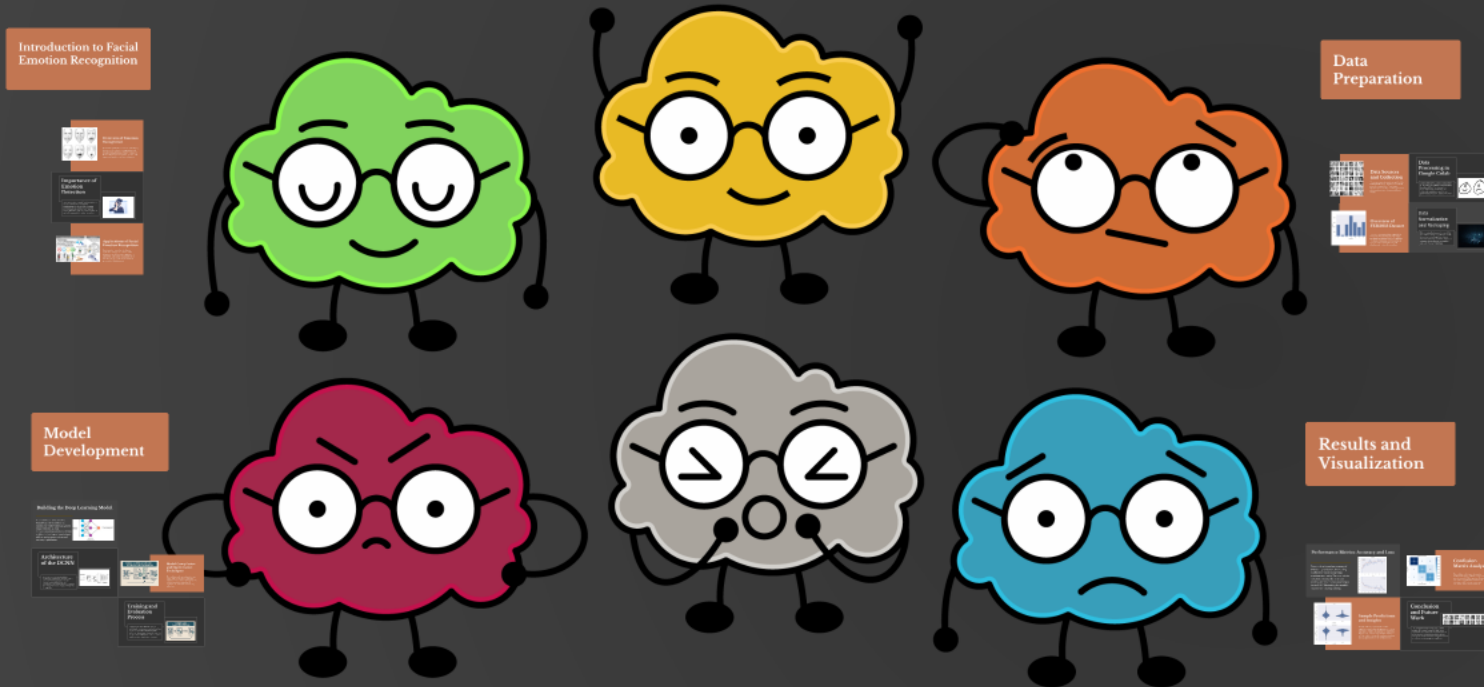
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# Model Development

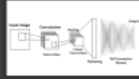
## Building the Deep Learning Model

The model is constructed using TensorFlow and Keras libraries, essential for implementing advanced neural networks. A Deep Convolutional Neural Network (DCNN) enables the training on large datasets while ensuring generalization and accuracy in prediction.



## Architecture of the DCNN

The architecture comprises multiple layers, convolutional layers for feature extraction, pooling layers for dimensionality reduction, and fully connected layers for classification. Data normalization and dropout techniques are implemented to enhance performance and reduce overfitting in the training process.



## MODEL COMPIATION AND OPTIMIZATION TECHNIQUES

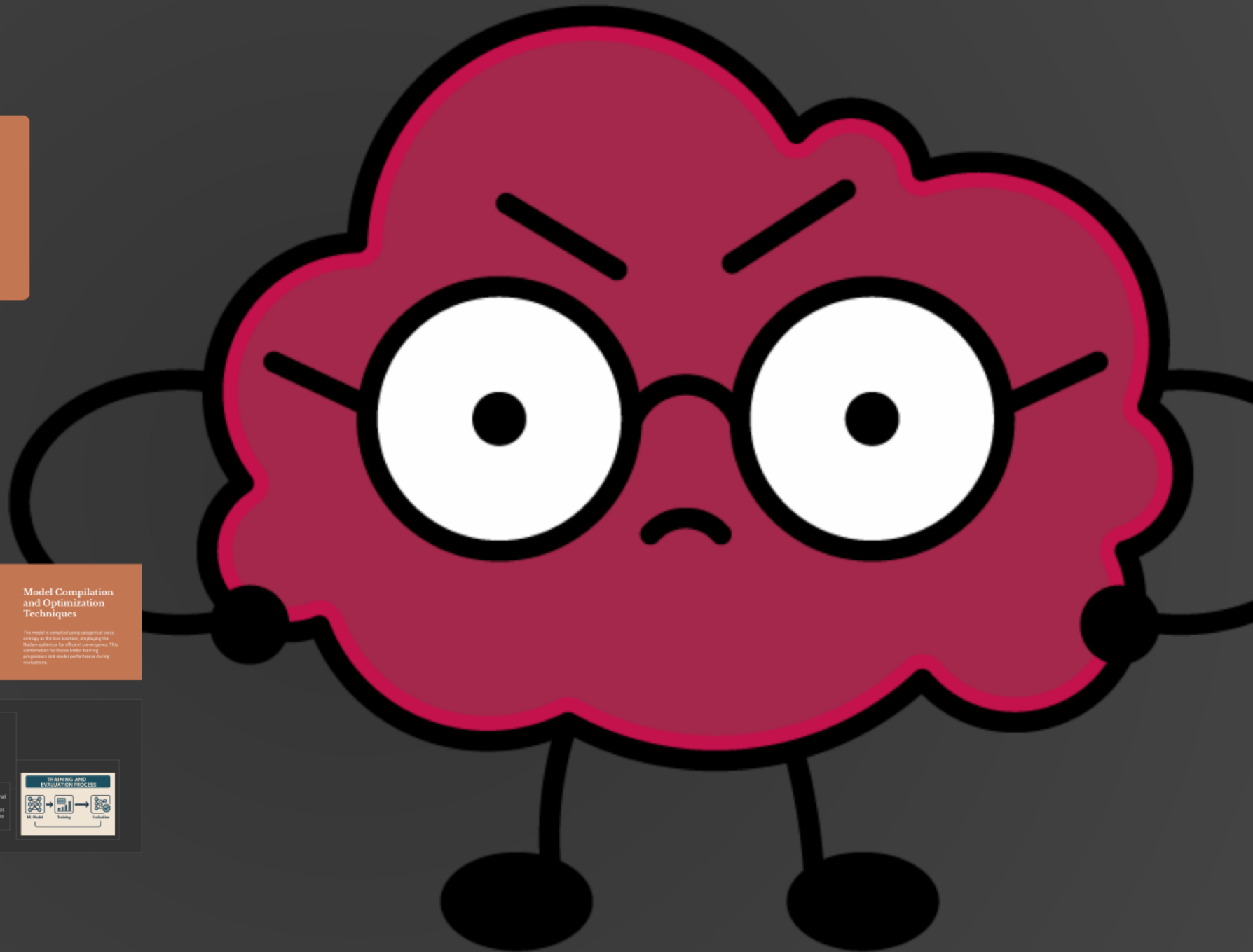


## Model Compilation and Optimization Techniques

The model is compiled using categorical cross-entropy as the loss function, ensuring the Keras optimizer for efficient convergence. This combination facilitates faster training progress and model performance during evaluations.

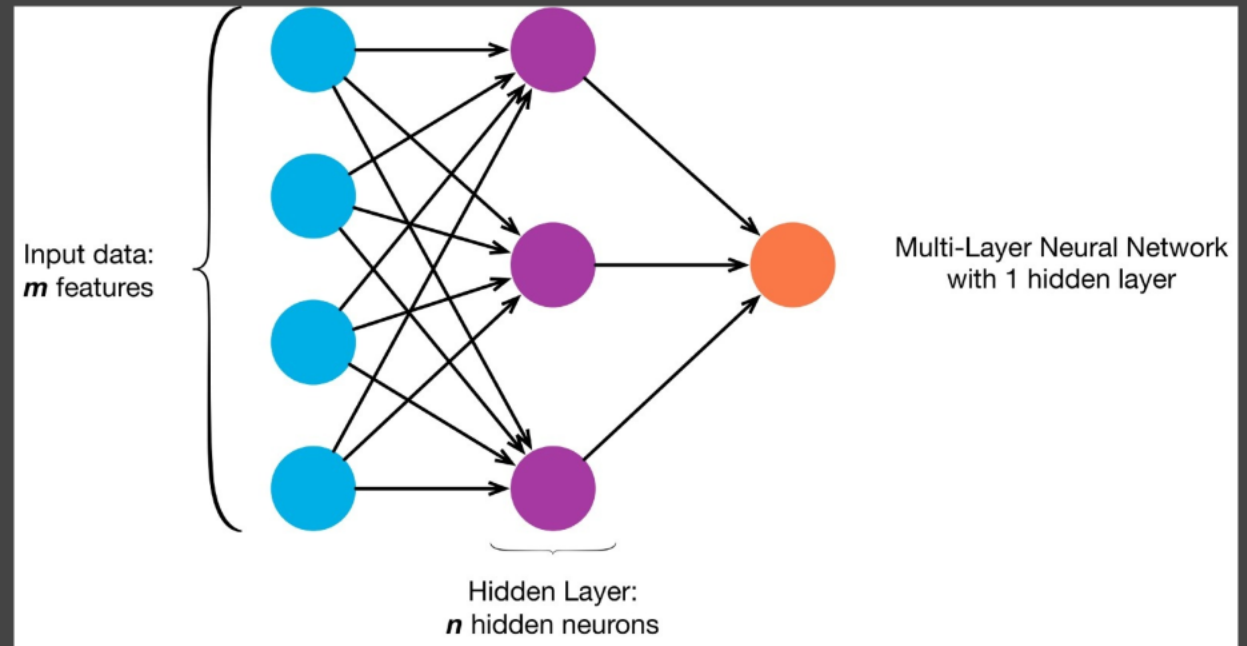
## Training and Evaluation Process

Training occurs with a dataset split into training and validation sets, aiming for optimal accuracy across 100 epochs. The model's performance is assessed using metrics such as accuracy, loss, and confusion matrix to ensure reliable emotion classification outcomes.



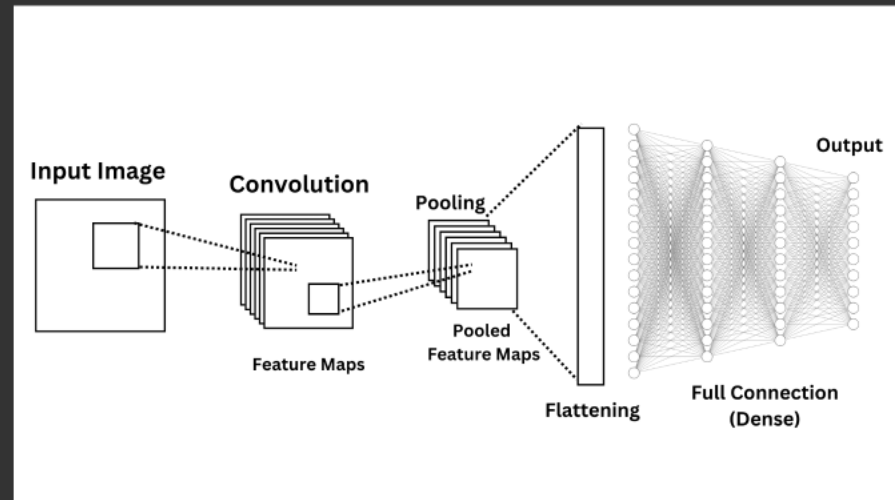
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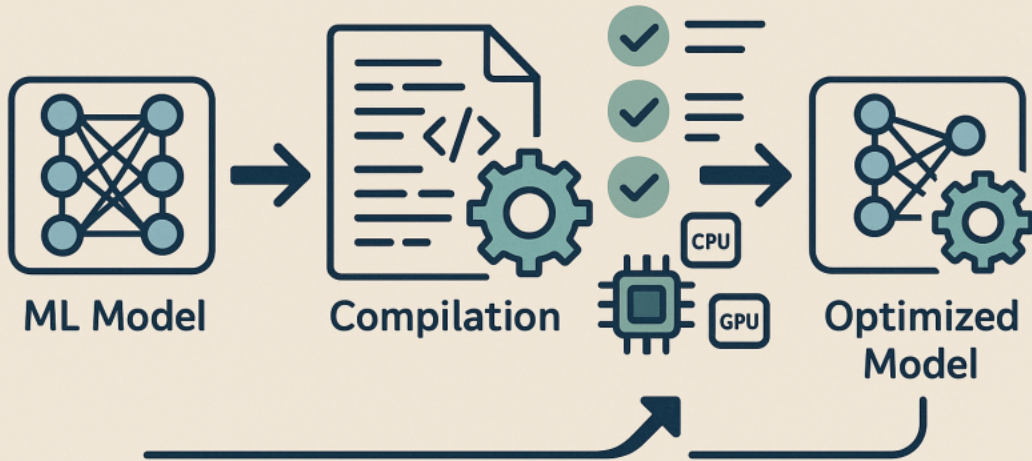
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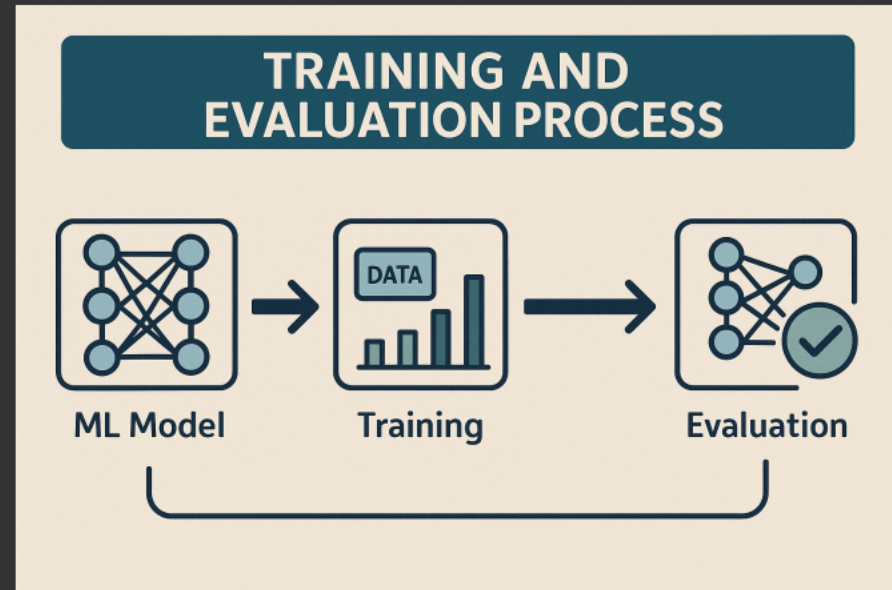


## Model Compilation and Optimization Techniques

The model is compiled using categorical cross-entropy as the loss function, employing the Nadam optimizer for efficient convergence. This combination facilitates better training progression and model performance during evaluations.

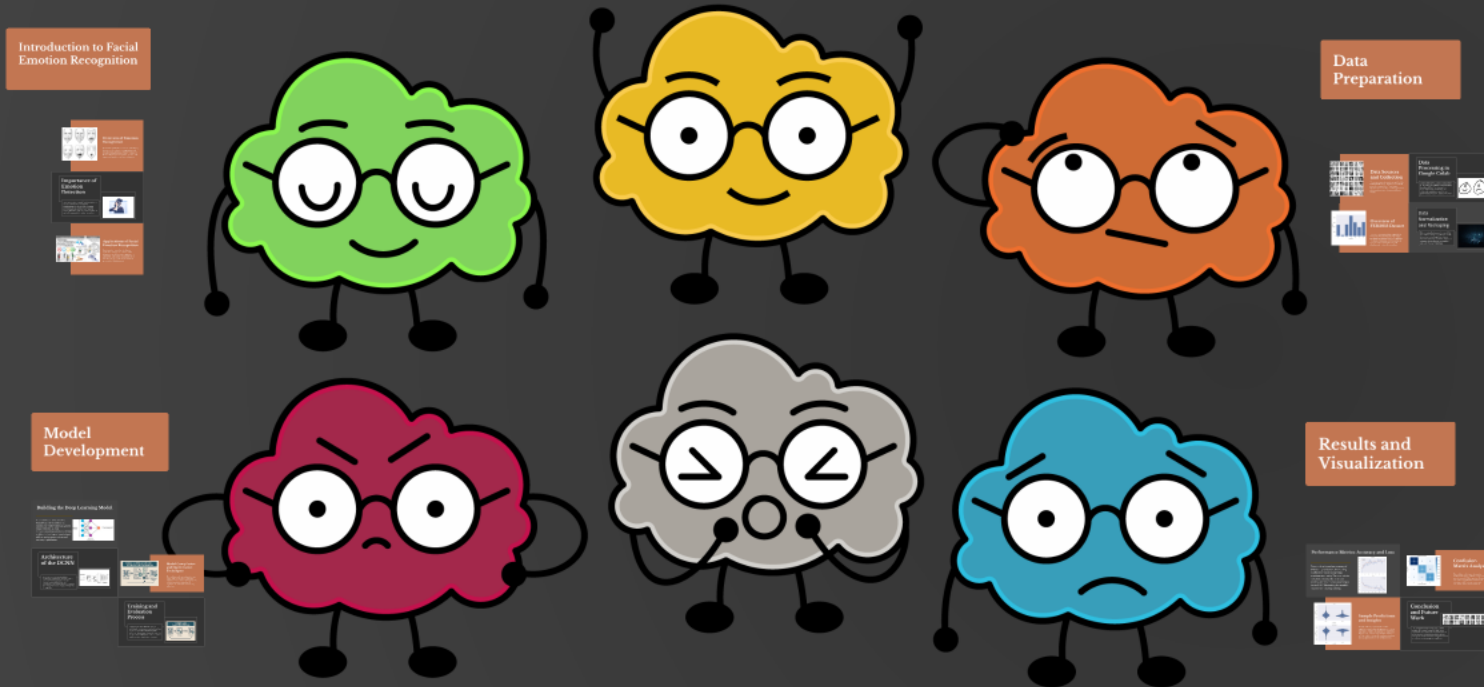
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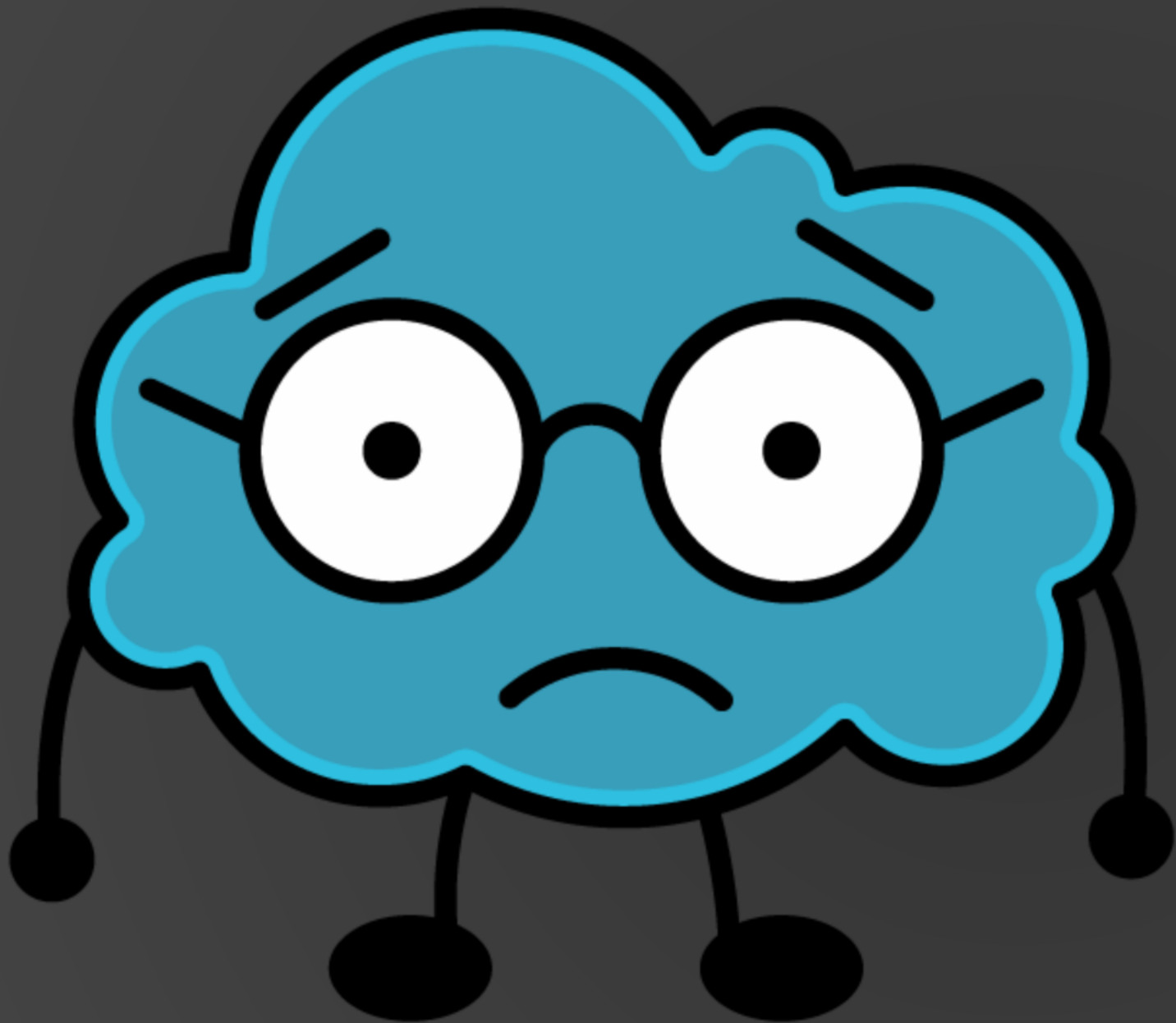


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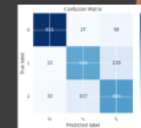
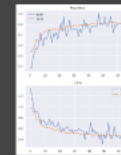
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# Results and Visualization

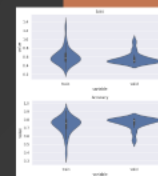
## Performance Metrics: Accuracy and Loss

The model achieved an accuracy of 82% during validation, showcasing its effectiveness in recognizing emotions accurately. The loss metrics indicated a steady decrease over epochs, with training loss reaching as low as 0.317, illustrating the model's improvement during training.



## Confusion Matrix Analysis

The confusion matrix displayed accurate predictions for 'happy' and 'sad' emotions, with a precision score of 0.95 for 'happy'. However, the model struggled with 'neutral' and 'surprise' emotions, indicating an area for improvement in its classification capabilities.



## Sample Predictions and Insights

Sample predictions revealed the model's capability to correctly identify emotions in varied expressions. Notably, the accuracy in recognizing 'happy' faces reinforced the dataset's robustness and the model's learning efficacy, demonstrating practical applications in real-world scenarios.

## Conclusion and Future Work

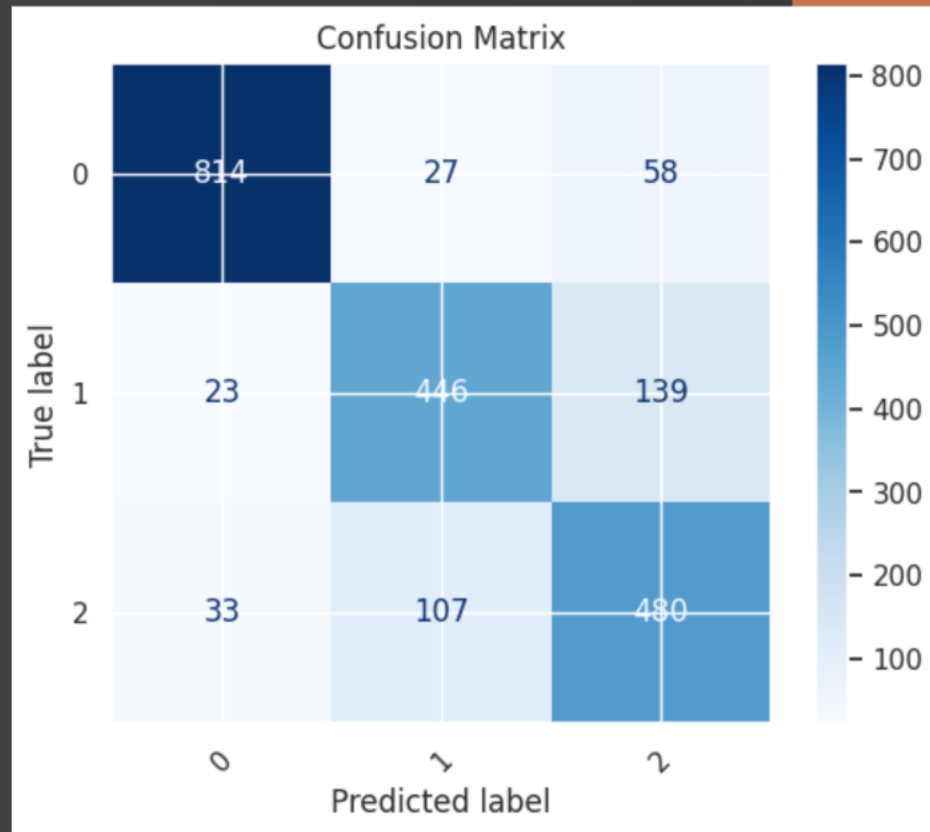
The strong performance metrics advocate for the model's robustness in emotion recognition tasks. Future work should address the misclassifications in neutral and surprise emotions through augmented training data and adjusted model parameters, aiming for improved accuracy and broader recognition capabilities.



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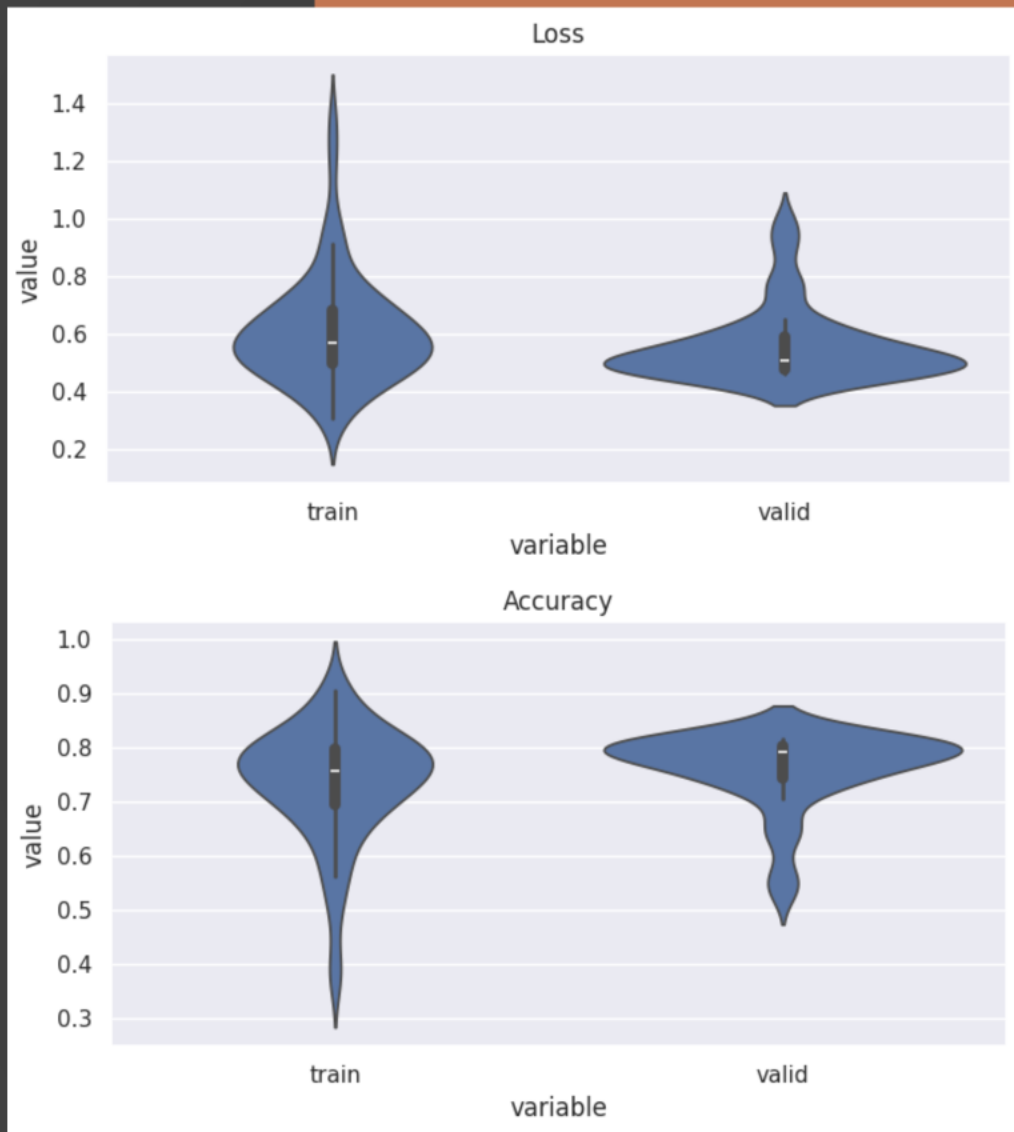
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# Confusion Matrix Analysis

The confusion matrix displayed accurate predictions for 'happy' and 'sad' emotions, with a precision score of 0.94 for happiness. However, the model struggled with neutral emotions, indicating an area for improvement, as it misclassified several of these instances.



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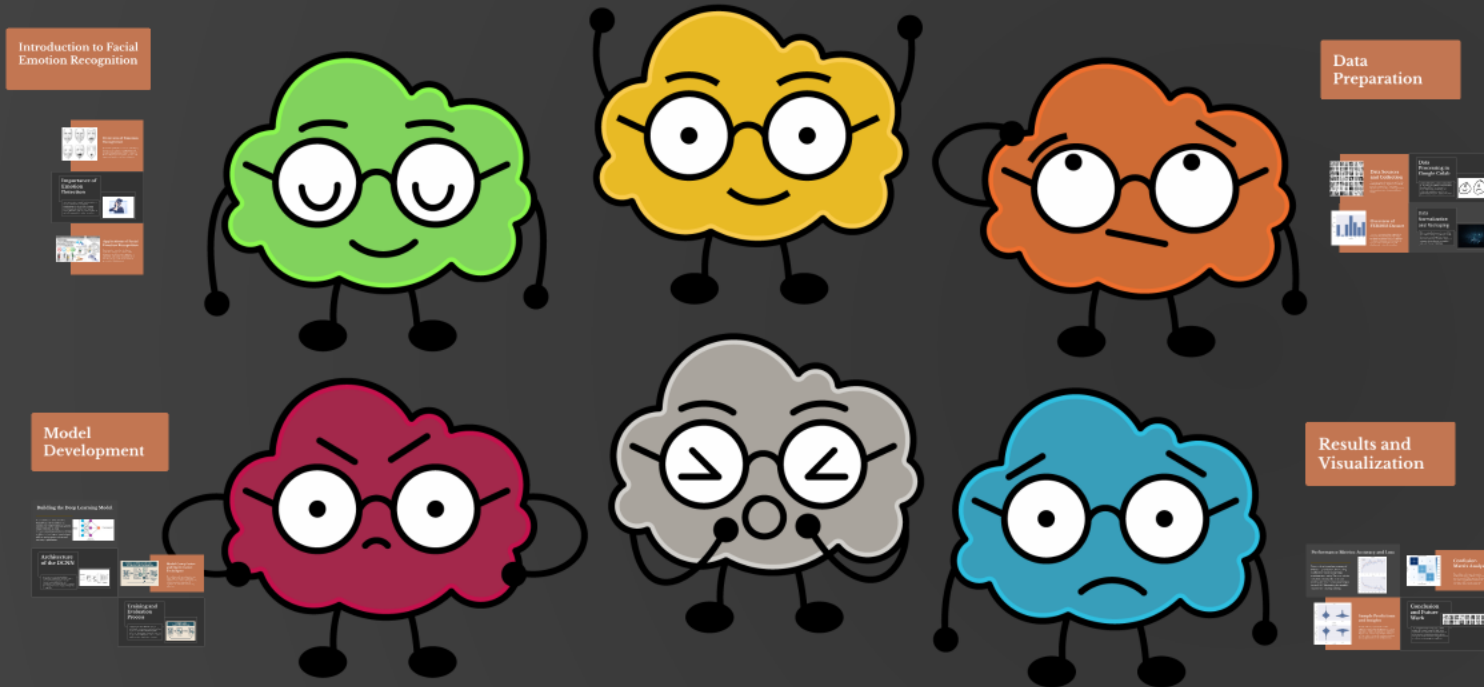
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